



INSTITUTO POLITÉCNICO NACIONAL

ESCUELA SUPERIOR DE CÓMPUTO

Materia: Procesamiento de Lenguaje Natural

Proyecto stage 01-Opinion Retrieval

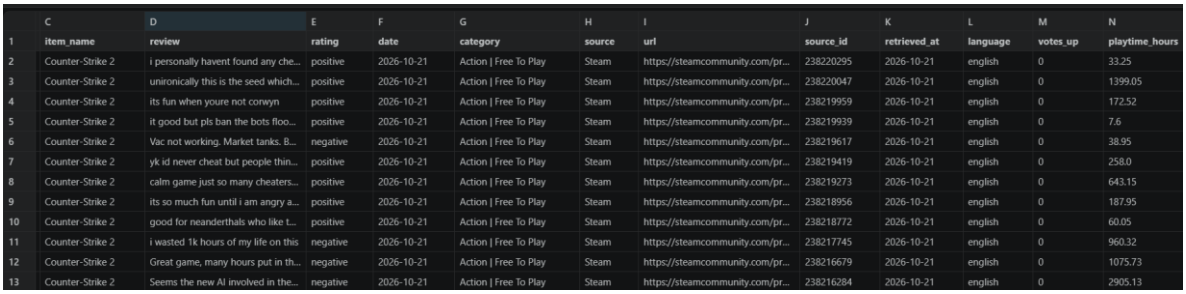
Grupo: 6AM1

Integrantes:

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- Cabello Cárdenas Ulises Abdiel
- Flores Bustos Karen
- Mexica Contreras Yaretzi

Resultados:

Estos fueron los resultados obtenidos de los datos obtenidos en donde se consulta una API de reseñas y los metadatos de cada juego obtiene su género/categoría real, verifica que los comentarios tengan texto y detecta si están en inglés. Descarta reseñas insuficientes y duplicadas.



	C	D	E	F	G	H	I	J	K	L	M	N
1	item_name	review	rating	date	category	source	url	source_id	retrieved_at	language	votes_up	playtime_hours
2	Counter-Strike 2	i personally havent found any che...	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238220295	2026-10-21	english	0	33.25
3	Counter-Strike 2	unironically this is the seed which...	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238220047	2026-10-21	english	0	1399.05
4	Counter-Strike 2	its fun when youre not corwyn	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238219959	2026-10-21	english	0	172.52
5	Counter-Strike 2	it good but pls ban the bots floo...	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238219939	2026-10-21	english	0	7.6
6	Counter-Strike 2	Vac not working. Market tanks. B...	negative	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238219617	2026-10-21	english	0	38.95
7	Counter-Strike 2	yk id never cheat but people thin...	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238219419	2026-10-21	english	0	258.0
8	Counter-Strike 2	calm game just so many cheaters...	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238219273	2026-10-21	english	0	643.15
9	Counter-Strike 2	its so much fun until i am angry a...	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238218956	2026-10-21	english	0	187.95
10	Counter-Strike 2	good for neanderthals who like t...	positive	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238218772	2026-10-21	english	0	60.05
11	Counter-Strike 2	i wasted 1k hours of my life on this	negative	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238217745	2026-10-21	english	0	960.32
12	Counter-Strike 2	Great game, many hours put in th...	negative	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238216679	2026-10-21	english	0	1075.73
13	Counter-Strike 2	Seems the new AI involved in the...	negative	2026-10-21	Action | Free To Play	Steam	https://steamcommunity.com/pr...	238216284	2026-10-21	english	0	2905.13

[Imagen 1: Opinions_raw_courpus]

En esta tabla se muestran de las reseñas anteriores obtenidas cada una se hizo la limpieza de la reseña aplicando los métodos necesarios y al igual se hizo la normalización de la reseña quitando stop words para realizarlo correctamente. Esto nos ayuda a lograr los siguientes pasos debidos para realizar el proyecto.



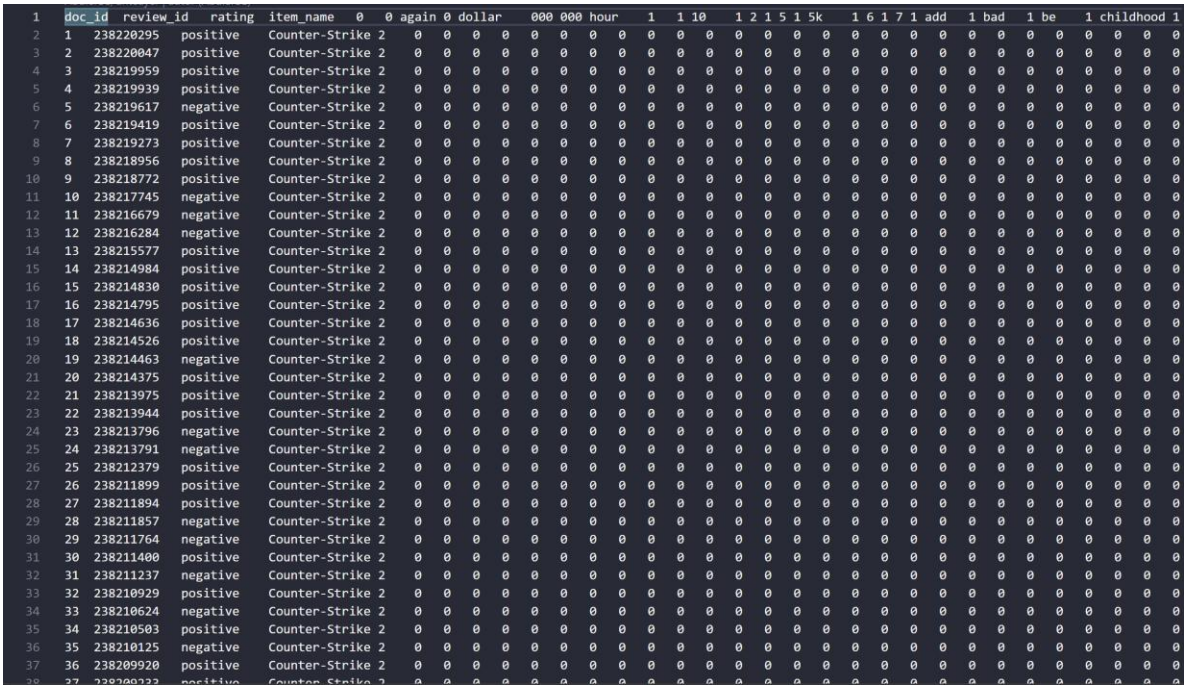
	I	J	K	L	M	N	O	P
1	url	source_id	retrieved_at	review	review_clean	review_norm	item_name_norm	item_review_norm
2	https://steamcommunity.com/pr...	238220295	2026-10-21	i personally havent found any che...	i personally haven't found any ch...	personally have not find any chea...	counter-strike 2	counter-strike 2 personall
3	https://steamcommunity.com/pr...	238220047	2026-10-21	unironically this is the seed which...	unironically this is the seed which...	unironically this be seed which sp...	counter-strike 2	counter-strike 2 unironica
4	https://steamcommunity.com/pr...	238219959	2026-10-21	its fun when youre not corwyn	its fun when youre not corwyn	fun when youre not corwyn	counter-strike 2	counter-strike 2 fun when
5	https://steamcommunity.com/pr...	238219939	2026-10-21	it good but pls ban the bots floo...	it good but pls ban the bots floo...	good pls ban bot flood comp cas...	counter-strike 2	counter-strike 2 good pls
6	https://steamcommunity.com/pr...	238219617	2026-10-21	Vac not working. Market tanks. B...	vac not working. market tanks. b...	vac not work market tank bot run...	counter-strike 2	counter-strike 2 vac not w
7	https://steamcommunity.com/pr...	238219419	2026-10-21	yk id never cheat but people thin...	yk id never cheat but people thin...	yk id never cheat people think do...	counter-strike 2	counter-strike 2 yk id neve
8	https://steamcommunity.com/pr...	238219273	2026-10-21	calm game just so many cheaters...	calm game just so many cheaters...	calm game just so many cheater ...	counter-strike 2	counter-strike 2 calm gam
9	https://steamcommunity.com/pr...	238218956	2026-10-21	its so much fun until i am angry a...	its so much fun until i am angry a...	so much fun be angry ahhahaha	counter-strike 2	counter-strike 2 so much f
10	https://steamcommunity.com/pr...	238218772	2026-10-21	good for neanderthals who like t...	good for neanderthals who like t...	good neanderthal like move cros...	counter-strike 2	counter-strike 2 good nea
11	https://steamcommunity.com/pr...	238217745	2026-10-21	i wasted 1k hours of my life on this	i wasted 1k hours of my life on this	waste 1k hour life this	counter-strike 2	counter-strike 2 waste 1k f
12	https://steamcommunity.com/pr...	238216679	2026-10-21	Great game, many hours put in th...	great game, many hours put in th...	great game many hour put game...	counter-strike 2	counter-strike 2 great gam
13	https://steamcommunity.com/pr...	238216284	2026-10-21	Seems the new AI involved in the...	seems the new ai involved in the ...	seem new ai involve vac sweep h...	counter-strike 2	counter-strike 2 seem new

[Imagen 2 : opinions_normalized_corpus]

A partir del corpus normalizado se construyeron las representaciones vectoriales de las reseñas. Cada fila corresponde a un documento (doc_id, review_id) con su calificación (rating) y el nombre del juego (item_name). El resto de las columnas

corresponde a los términos del vocabulario. Se generaron tres tipos de representación (binaria, de frecuencia y TF-IDF), cada una en dos variantes: solo con el texto de la reseña, y con el nombre del juego concatenado a la reseña.

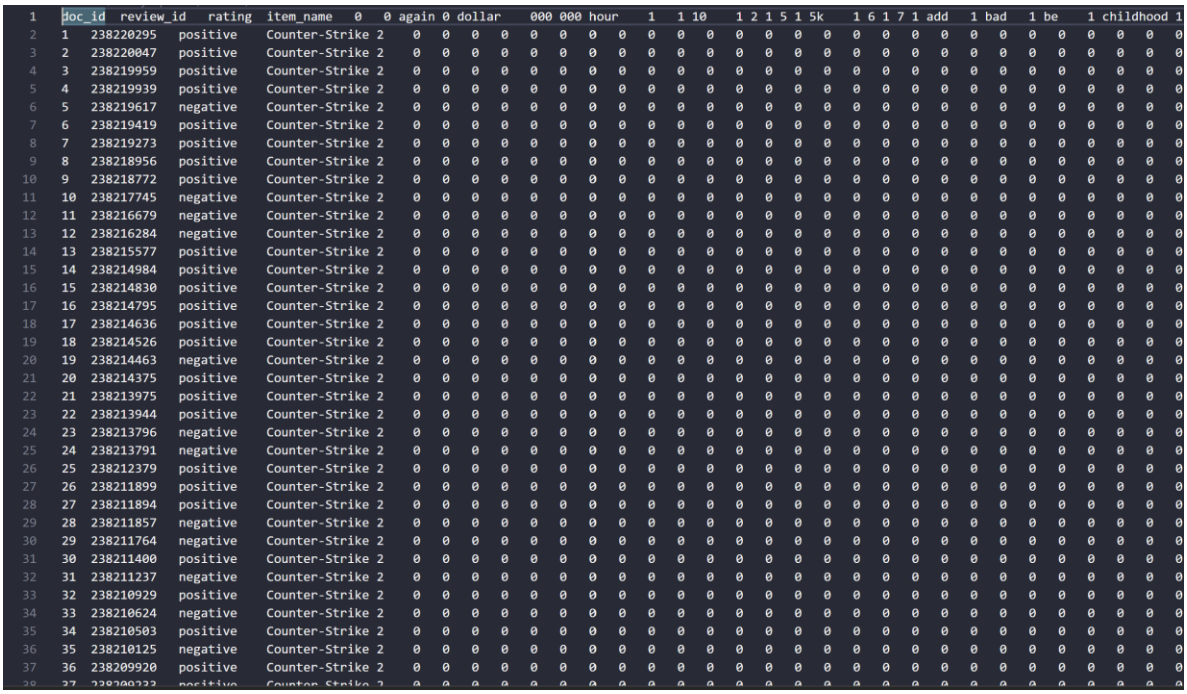
En esta matriz se representa cada reseña de forma binaria usando únicamente el texto de la reseña. Los ceros que se ven en la captura corresponden a los primeros términos del vocabulario, que están en orden alfabético y son tokens numéricos poco frecuentes. Por eso la matriz es dispersa.



	doc_id	review_id	rating	item_name	0	0 again	0 dollar	000	000 hour	1	1 10	1 2 1 5 1 5k	1 6 1 7 1 add	1 bad	1 be	1 childhood	1
1	1	238220295	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
2	2	238220047	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
3	3	238219959	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
4	4	238219939	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
5	5	238219617	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
6	6	238219419	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
7	7	238219273	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
8	8	238218956	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
9	9	238218772	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
10	10	238217745	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
11	11	238216679	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
12	12	238216284	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
13	13	238215577	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
14	14	238214984	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
15	15	238214830	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
16	16	238214795	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
17	17	238214636	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
18	18	238214526	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
19	19	238214463	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
20	20	238214375	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
21	21	238213975	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
22	22	238213944	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
23	23	238213796	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
24	24	238213791	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
25	25	238212379	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
26	26	238211899	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
27	27	238211894	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
28	28	238211857	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
29	29	238211764	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
30	30	238211400	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
31	31	238211237	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
32	32	238210929	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
33	33	238210624	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
34	34	238210503	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
35	35	238210125	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
36	36	238209920	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0

[Imagen 3: `vectores_review_only_binary`*]*

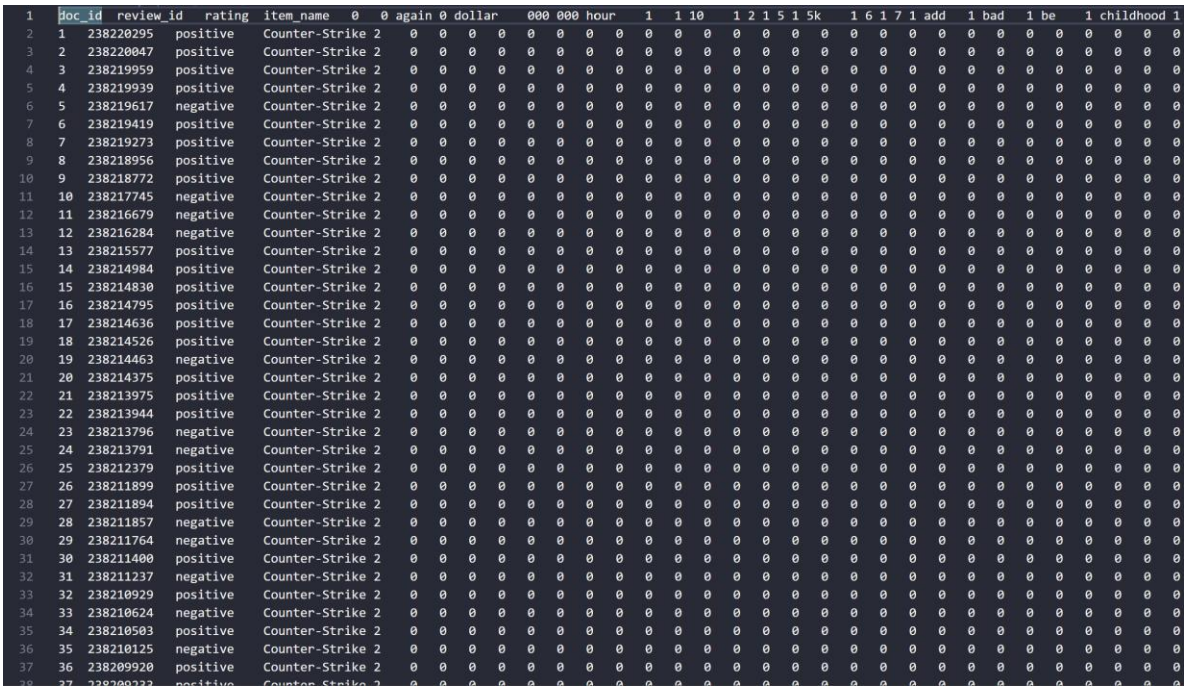
Esta matriz sigue la misma lógica binaria, pero el texto de cada documento es la concatenación del nombre del juego con la reseña (item_name + review). Así, los términos del título también cuentan como parte del documento y pueden coincidir con las consultas. Esto permite recuperar reseñas por el nombre del juego además de por su contenido.



doc_id	review_id	rating	item_name	0	0 again	0 dollar	000	000 hour	1	1 10	1 2 1 5 1 5k	1 6 1 7 1 add	1 bad	1 be	1 childhood	1
1	238220295	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
2	238220047	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
3	238219959	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
4	238219939	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
5	238219617	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
6	238219419	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
7	238219273	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
8	238218956	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
9	238218772	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
10	238217745	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
11	238216679	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
12	238216284	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
13	238215577	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
14	238214984	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
15	238214830	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
16	238214795	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
17	238214636	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
18	238214526	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
19	238214463	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
20	238214375	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
21	238213975	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
22	238213944	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
23	238213796	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
24	238213791	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
25	238212379	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
26	238211899	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
27	238211894	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
28	238211857	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
29	238211764	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
30	238211400	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
31	238211237	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
32	238210929	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
33	238210624	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
34	238210503	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
35	238210125	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
36	238209920	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
37	238209333	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0

[Imagen 4: `vectores_item_name_plus_review_binary`*]*

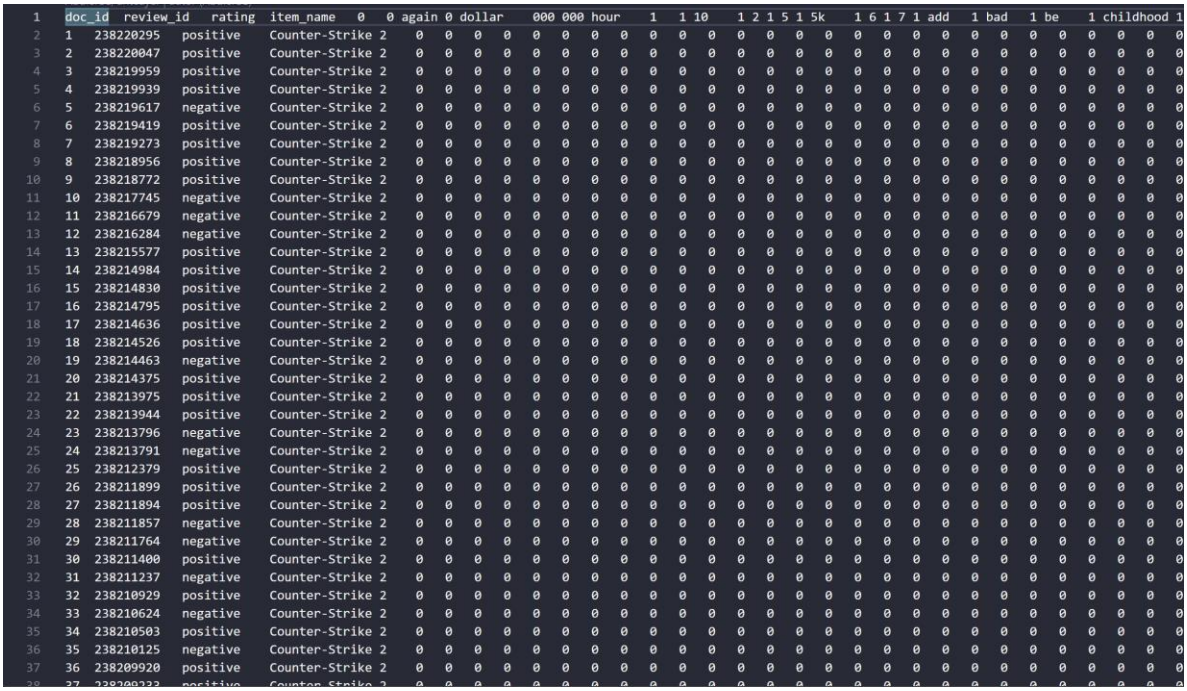
En esta matriz cada valor es el número de veces que un término aparece en la reseña, a diferencia de la representación binaria, que solo indica presencia. Esto da más peso a los términos que el usuario repite. En la captura predominan los ceros porque se muestran los primeros términos del vocabulario, que casi no aparecen en las reseñas.



doc_id	review_id	rating	item_name	0	0 again	0 dollar	000	000 hour	1	1 10	1 2 1 5 1 5k	1 6 1 7 1 add	1 bad	1 be	1 childhood	1
1	238220295	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
2	238220047	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
3	238219959	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
4	238219939	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
5	238219617	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
6	238219419	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
7	238219273	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
8	238218956	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
9	238218772	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
10	238217745	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
11	238216679	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
12	238216284	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
13	238215577	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
14	238214984	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
15	238214830	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
16	238214795	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
17	238214636	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
18	238214526	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
19	238214463	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
20	238214375	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
21	238213975	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
22	238213944	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
23	238213796	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
24	238213791	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
25	238212379	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
26	238211899	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
27	238211894	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
28	238211857	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
29	238211764	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
30	238211400	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
31	238211237	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
32	238210929	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
33	238210624	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
34	238210503	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
35	238210125	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
36	238209920	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0

[Imagen 5: `vectores_review_only_frequency`*]*

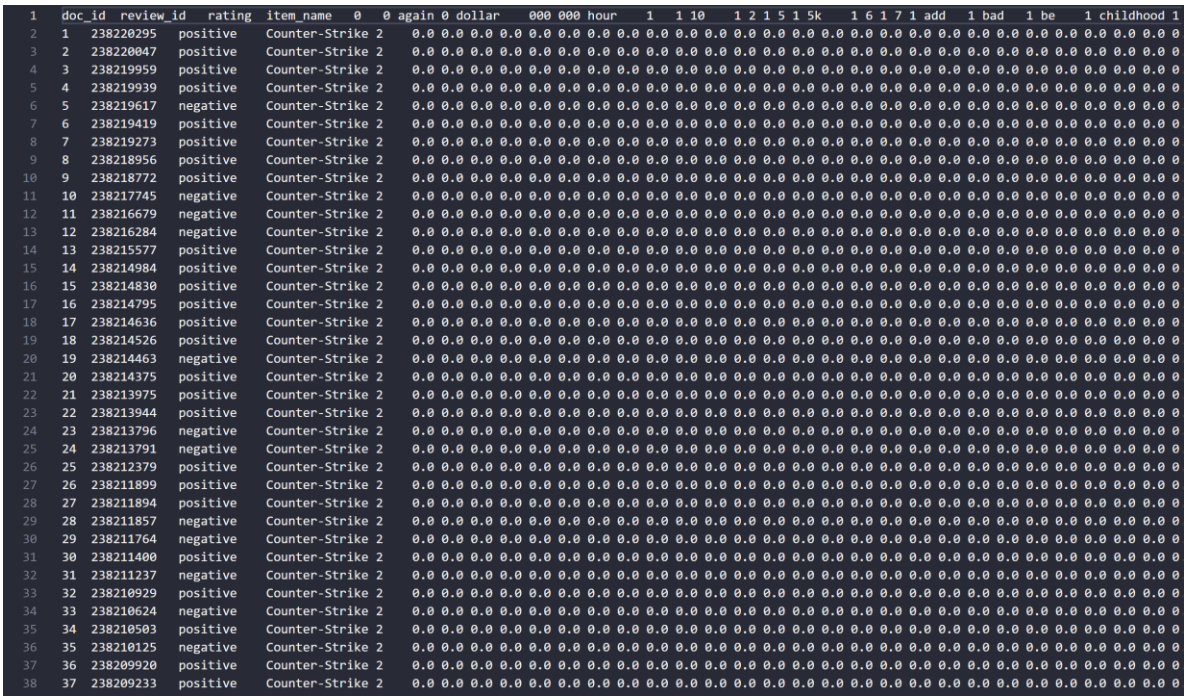
Esta matriz es la versión por frecuencia con el nombre del juego concatenado a la reseña. El conteo de términos incluye también las palabras del título, de modo que el nombre del juego aporta peso a la representación de cada documento.



doc_id	review_id	rating	item_name	0	0 again	0 dollar	000	000 hour	1	1 10	1 2 1 5 1 5k	1 6 1 7 1 add	1 bad	1 be	1 childhood	1
1	238220295	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
2	238220047	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
3	238219959	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
4	238219939	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
5	238219617	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
6	238219419	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
7	238219273	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
8	238218956	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
9	238218772	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
10	238217745	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
11	238216679	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
12	238216284	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
13	238215577	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
14	238214984	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
15	238214830	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
16	238214795	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
17	238214636	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
18	238214526	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
19	238214463	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
20	238214375	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
21	238213975	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
22	238213944	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
23	238213796	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
24	238213791	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
25	238212379	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
26	238211899	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
27	238211894	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
28	238211857	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
29	238211764	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
30	238211400	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
31	238211237	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
32	238210929	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
33	238210624	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
34	238210503	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
35	238210125	negative	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0
36	238209920	positive	Counter-Strike 2	0	0	0	0	0	0	0	0	0	0	0	0	0

[Imagen 6: `vectores_item_name_plus_review_frequency`*]*

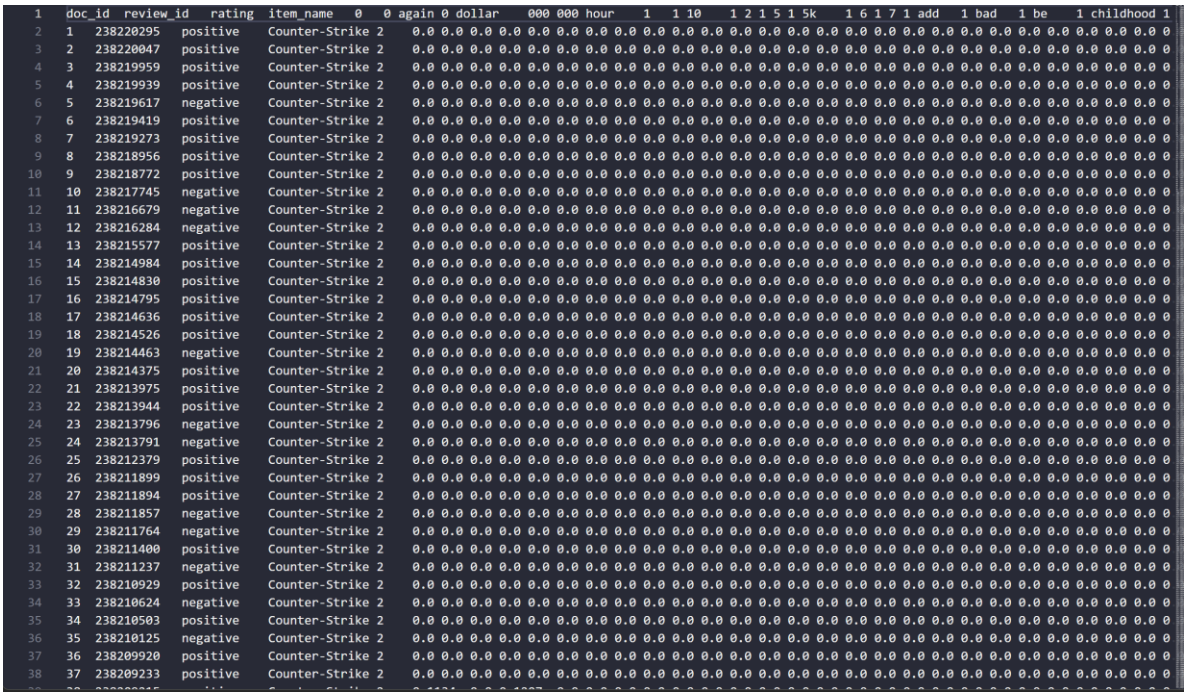
En esta matriz cada valor es el peso TF-IDF del término en la reseña. Los términos muy comunes reciben un peso bajo y los más distintivos uno mayor. Los valores 0.0 de la captura indican que esos términos no aparecen en las reseñas mostradas.



	doc_id	review_id	rating	item_name	0	0 again	0 dollar	000	000 hour	1	1 10	1 2	1 5	1 5k	1 6	1 7	1 add	1 bad	1 be	1 childhood	1
1	1	238220295	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
2	2	238220047	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
3	3	238219959	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
4	4	238219939	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
5	5	238219617	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
6	6	238219419	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
7	7	238219273	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
8	8	238218956	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
9	9	238218772	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
10	10	238217745	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
11	11	238216679	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
12	12	238216284	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
13	13	238215577	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
14	14	238214984	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
15	15	238214830	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
16	16	238214795	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
17	17	238214636	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
18	18	238214526	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
19	19	238214463	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
20	20	238214375	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
21	21	238213975	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
22	22	238213944	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
23	23	238213796	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
24	24	238213791	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
25	25	238212379	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
26	26	238211899	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
27	27	238211894	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
28	28	238211857	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
29	29	238211764	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
30	30	238211400	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
31	31	238211237	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
32	32	238210929	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
33	33	238210624	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
34	34	238210503	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
35	35	238210125	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
36	36	238209920	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
37	37	238209233	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0

[Imagen 7: `vectores_review_only_tfidf`*]*

Esta matriz es la versión TF-IDF con el nombre del juego concatenado a la reseña. Al incluir el título en el texto, sus términos también se ponderan por su rareza en el corpus. Esto puede dar más importancia a los nombres de los juegos, que aparecen en pocos documentos distintos.



doc_id	review_id	rating	item_name	0	0 again	0 dollar	000	000 hour	1	1 10	1 2 1 5 1 5k	1 6 1 7 1 add	1 bad	1 be	1 childhood	1
1	238220295	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
2	238220047	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
3	238219959	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
4	238219939	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
5	238219617	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
6	238219419	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
7	238219273	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
8	238218956	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
9	238218772	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
10	238217745	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
11	238216679	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
12	238216284	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
13	238215577	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
14	238214984	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
15	238214830	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
16	238214795	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
17	238214636	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
18	238214526	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
19	238214463	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
20	238214375	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
21	238213975	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
22	238213944	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
23	238213796	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
24	238213791	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
25	238212379	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
26	238211899	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
27	238211894	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
28	238211857	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
29	238211764	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
30	238211400	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
31	238211237	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
32	238210929	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
33	238210624	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
34	238210503	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
35	238210125	negative	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
36	238209920	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
37	238209233	positive	Counter-Strike 2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0

[Imagen 8: `vectores_item_name_plus_review_tfidf`*]*

Para evaluar el sistema de recuperación se definió un conjunto de 15 consultas (Q01 a Q15), cada una con un identificador (query_id), un nivel de dificultad (difficulty_level) y el texto de la consulta (query). Las consultas del nivel 1 son simples y directas, como “cheaters in competitive matches”. Las del nivel 2 describen situaciones más específicas, y las del nivel 3 expresan ideas con más contexto, como “the world rewards curiosity and exploration”.

	query_id	difficulty_level	query
1	Q01	1	cheaters in competitive matches
2	Q02	1	game crashes and freezes
3	Q03	1	great story and characters
4	Q04	1	difficult boss fights
5	Q05	1	poor server connection
6	Q06	2	the game runs badly on my computer
7	Q07	2	playing with friends makes the experience better
8	Q08	2	too many paid items and microtransactions
9	Q09	2	the controls are frustrating to learn
10	Q10	2	repetitive activities after many hours
11	Q11	3	unfair opponents ruin online matches
12	Q12	3	technical problems interrupt the experience
13	Q13	3	the world rewards curiosity and exploration
14	Q14	3	the game demands a lot of practice to improve
15	Q15	3	the experience becomes boring over time

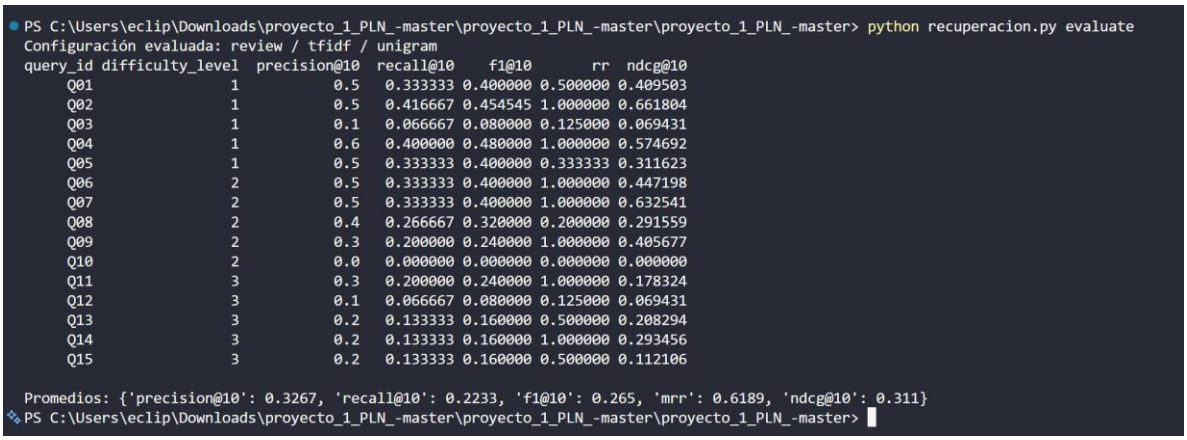
[Imagen 9: `Conjunto de consultas de evaluación por nivel de dificultad`]

Para cada consulta, el sistema devuelve un ranking de los 10 documentos más similares. La tabla muestra el identificador de la consulta (query_id), el documento recuperado (document_id), su posición (rank) y su puntuación de similitud (score). Por ejemplo, en Q01 los puntajes bajan desde 0.298 en el primer lugar. En Q02 hay una diferencia grande entre el primer documento (0.280) y el segundo (0.145), lo que indica que solo un documento coincide con la consulta.

	query_id	document_id	rank	score
2	Q01	36	1	0.2981601807430715
3	Q01	37	2	0.26171534246902495
4	Q01	34	3	0.25478417417504506
5	Q01	59	4	0.25294070748642966
6	Q01	38	5	0.2420346457222375
7	Q01	186	6	0.23393230431466844
8	Q01	1	7	0.22824553023639724
9	Q01	80	8	0.22095579770896476
10	Q01	89	9	0.21496362398573668
11	Q01	152	10	0.2118457712956903
12	Q02	106	1	0.2803732374487358
13	Q02	5	2	0.1448626543530396
14	Q02	456	3	0.11046001430349169
15	Q02	257	4	0.09955717719412588
16	Q02	165	5	0.07630281175838907
17	Q02	192	6	0.06640609368956153
18	Q02	482	7	0.06197648546911171

[Imagen 10: `Ranking de documentos recuperados por consulta (query_id, document_id, rank, score)`]

Se ejecutó la evaluación con la configuración review / tfidf / unigram y se calcularon por consulta las métricas precision, recall, F1, rr (rango recíproco) . En promedio se obtuvo una precisión de 0.3267, un recal de 0.2233, un F1 de 0.265, un MRR de 0.6189 y un nDCG de 0.311. El MRR es la métrica más alta, así que el primer documento relevante suele aparecer en las primeras posiciones, pero hay pocos relevantes en el top 10. El desempeño baja con la dificultad: la precisión promedio es 0.44 en el nivel 1, 0.34 en el nivel 2 y 0.20 en el nivel 3. Q02 y Q07 tienen los mejores nDCG (0.66 y 0.63), mientras que Q10 (“repetitive activities after many hours”) obtuvo 0 en todas las métricas.



[Imagen 11: Evaluación del sistema (review, tfidf, unigram) con métricas por consulta y promedios]

La siguiente tabla resume el desempeño global del sistema sobre las 15 consulta. Aproximadamente 3 de cada 10 documentos recuperados en el top 10 son relevantes. El MRR es la métrica más alta, lo que indica que el primer documento relevante suele aparecer en las primeras posiciones del ranking.

1	queries	precision@10	recall@10	f1@10	mrr	ndcg@10
2	15	0.3266666666666666	0.22333333333333336	0.264969696969697	0.6188888888888889	0.31104262023598167

[Imagen 12: Promedios globales de las métricas de evaluación

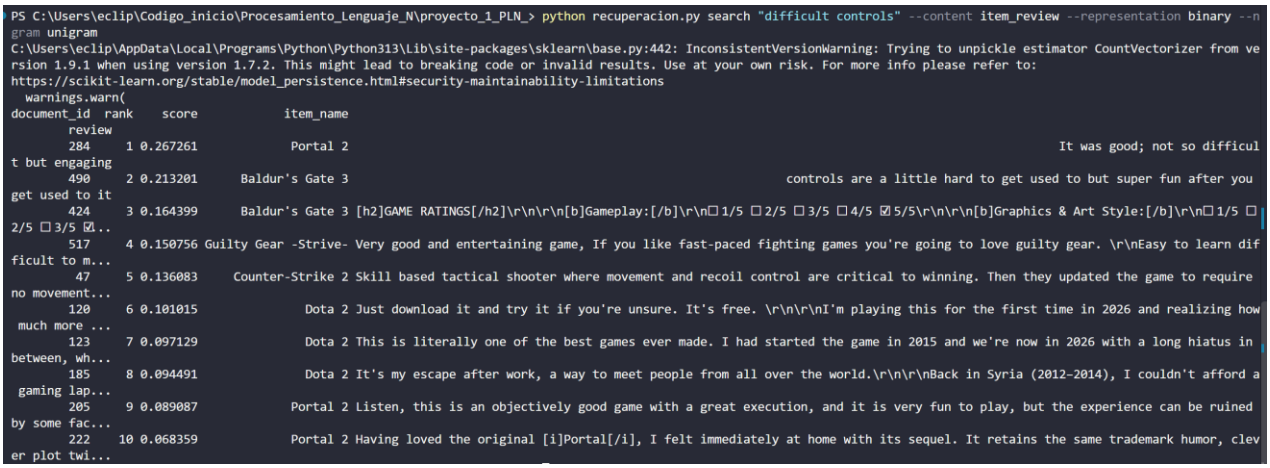
Análisis de Errores:

Para el análisis de errores una vez tenemos los siguientes queries con good perfomance and poor perfomance

Good Perfomance

1) python recuperacion.py search "difficult controls" --content item_review --representation binary --ngram unigram

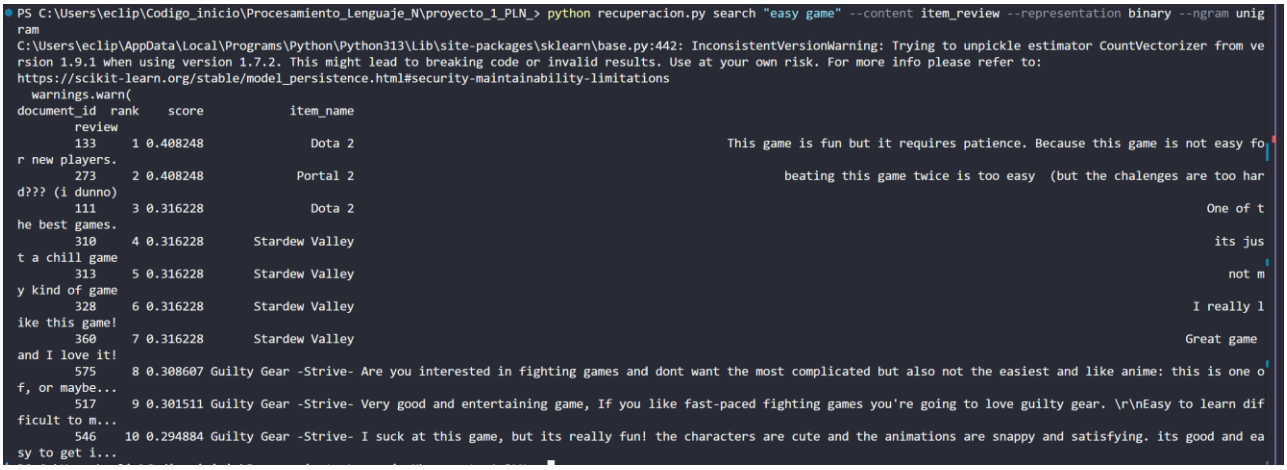
Con esta consulta obtuvimos buenos resultados ya que tenemos palabras similares, pero también hubo reviews que arrojaron expresiones regulares, y en si la similitud, es más referente al contexto y significado de las palabras.



[Imagen 13: Resultados de "difficult controls" (binary, unigram, item_review)]

2) **python recuperacion.py search "easy game" --content item_review --representation binary --ngram unigram**

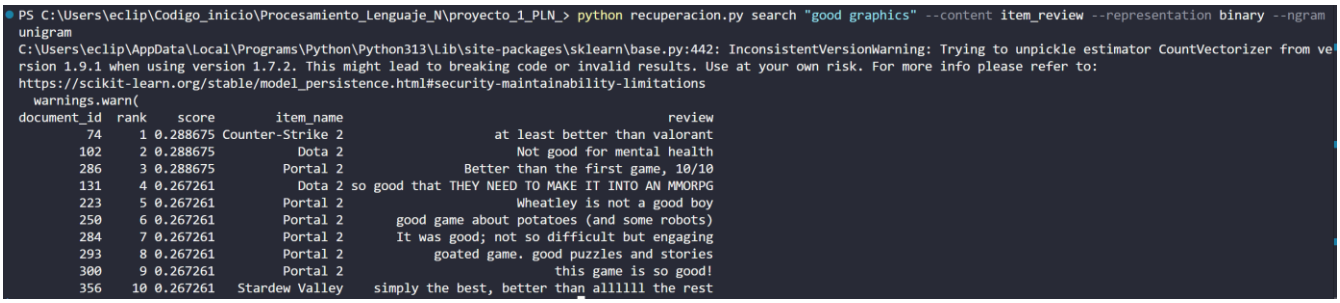
Para esta consulta obtuvimos resultados con palabras similares o la misma palabra, pero con contextos variados que se podría interpretar como que los resultados son tomados dependiendo de una de las dos palabras usadas probablemente porque son unigramas.



[Imagen 14: Resultados de "easy game" (binary, unigram, item_review)]

3) **python recuperacion.py search "good graphics" --content item_review --representation binary --ngram unigram**

Se obtuvieron buenos resultados en cuanto a obtener palabras que son sinónimos de good, pero como nota los resultados no reflejan la parte de graphics lo cual podria deberse a que como buscamos por unigramas tomo solo good al realizar la búsqueda y al encontrar buenos resultados se detuvo ahi.

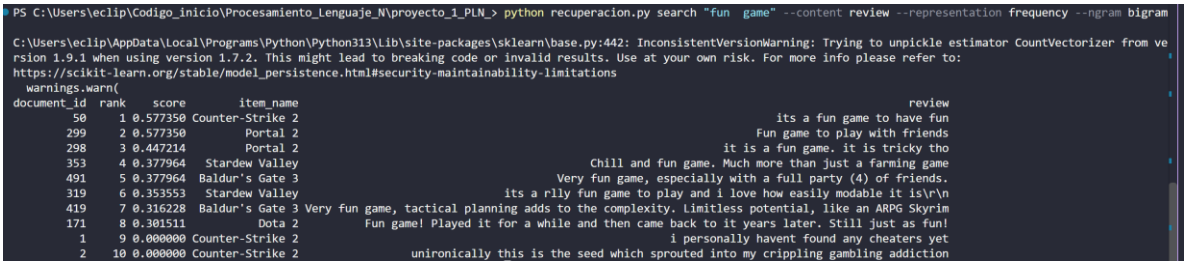


```
PS C:\Users\eclip\Codigo_inicio\Procesamiento_Lenguaje_N\proyecto_1_PLN_> python recuperacion.py search "good graphics" --content item_review --representation binary --ngram
unigram
C:\Users\eclip\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\base.py:442: InconsistentVersionWarning: Trying to unpickle estimator CountVectorizer from ve
rsion 1.9.1 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
document_id  rank    score        item_name                                        review
         74     1 0.288675 Counter-Strike 2                  at least better than valorant
        102     2 0.288675           Dota 2                      Not good for mental health
        286     3 0.288675         Portal 2                Better than the first game, 10/10
        131     4 0.267261           Dota 2 so good that THEY NEED TO MAKE IT INTO AN MMORPG
        223     5 0.267261         Portal 2                      Wheatley is not a good boy
        250     6 0.267261         Portal 2       good game about potatoes (and some robots)
        284     7 0.267261         Portal 2       It was good; not so difficult but engaging
        293     8 0.267261         Portal 2            goated game. good puzzles and stories
        300     9 0.267261         Portal 2                        this game is so good!
        356    10 0.267261   Stardew Valley    simply the best, better than allllll the rest
```

[Imagen 15: Resultados de "good graphics" (binary, unigram, item_review)]

4)**python recuperacion.py search "fun game" --content review --representation frequency --ngram bigram**

Para esta consulta parece ser que si son las palabras exactas contenidas dentro del review, el bigrama funciona bien arrojando resultados que contienen las dos palabras usadas en la consulta.



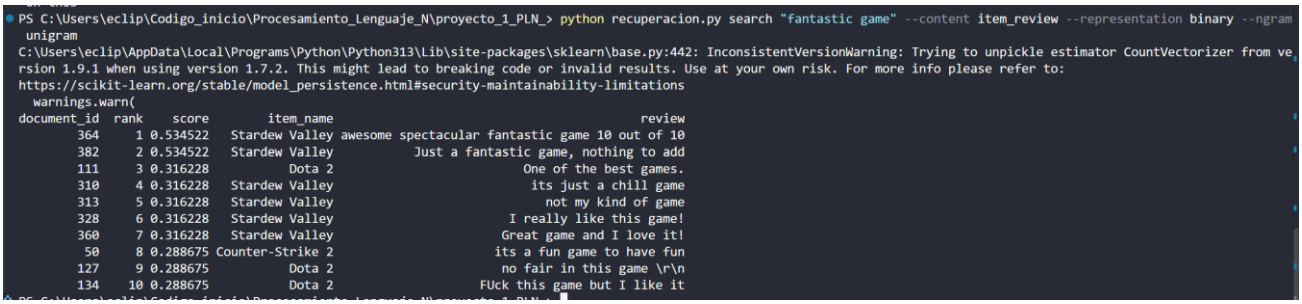
```
PS C:\Users\eclip\Codigo_inicio\Procesamiento_Lenguaje_N\proyecto_1_PLN_> python recuperacion.py search "fun  game" --content review --representation frequency --ngram bigram

C:\Users\eclip\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\base.py:442: InconsistentVersionWarning: Trying to unpickle estimator CountVectorizer from ve
rsion 1.9.1 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
document_id  rank    score        item_name                                                                                                review
         50     1 0.577350 Counter-Strike 2                                                                                 its a fun game to have fun
        299     2 0.577350         Portal 2                                                                                 Fun game to play with friends
        298     3 0.447214         Portal 2                                                                              it is a fun game. it is tricky tho
        353     4 0.377964   Stardew Valley                                                                  Chill and fun game. Much more than just a farming game
        491     5 0.377964   Baldur's Gate 3                                                          Very fun game, especially with a full party (4) of friends.
        319     6 0.353553   Stardew Valley                                                 its a rlly fun game to play and i love how easily modable it is\r\n
        419     7 0.316228   Baldur's Gate 3 Very fun game, tactical planning adds to the complexity. Limitless potential, like an ARPG Skyrim
        171     8 0.301511           Dota 2         Fun game! Played it for a while and then came back to it years later. Still just as fun!
          1     9 0.000000 Counter-Strike 2                                                                          i personally havent found any cheaters yet
          2    10 0.000000 Counter-Strike 2                unironically this is the seed which sprouted into my crippling gambling addiction
```

[Imagen 16: Resultados de "fun game" (frequency, bigram, review)]

5) **python recuperacion.py search "fantastic game" --content item_review --representation binary --ngram unigram**

Para esta consulta obtuvimos resultados con palabras similares, con contexto variado, donde vemos que al menos una de las palabras está presente o palabras similares.



```
PS C:\Users\eclip\Codigo_inicio\Procesamiento_Lenguaje_N\proyecto_1_PLN_> python recuperacion.py search "fantastic game" --content item_review --representation binary --ngram
  unigram
C:\Users\eclip\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\base.py:442: InconsistentVersionWarning: Trying to unpickle estimator CountVectorizer from ve
rsion 1.9.1 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
document_id  rank    score       item_name                                    review
        364     1 0.534522   Stardew Valley awesome spectacular fantastic game 10 out of 10
        382     2 0.534522   Stardew Valley          Just a fantastic game, nothing to add
        111     3 0.316228          Dota 2                        One of the best games.
        310     4 0.316228   Stardew Valley                         its just a chill game
        313     5 0.316228   Stardew Valley                           not my kind of game
        328     6 0.316228   Stardew Valley                      I really like this game!
        360     7 0.316228   Stardew Valley                     Great game and I love it!
         50     8 0.288675 Counter-Strike 2                    its a fun game to have fun
        127     9 0.288675          Dota 2                     no fair in this game \r\n
        134    10 0.288675          Dota 2                   FUck this game but I like it
```

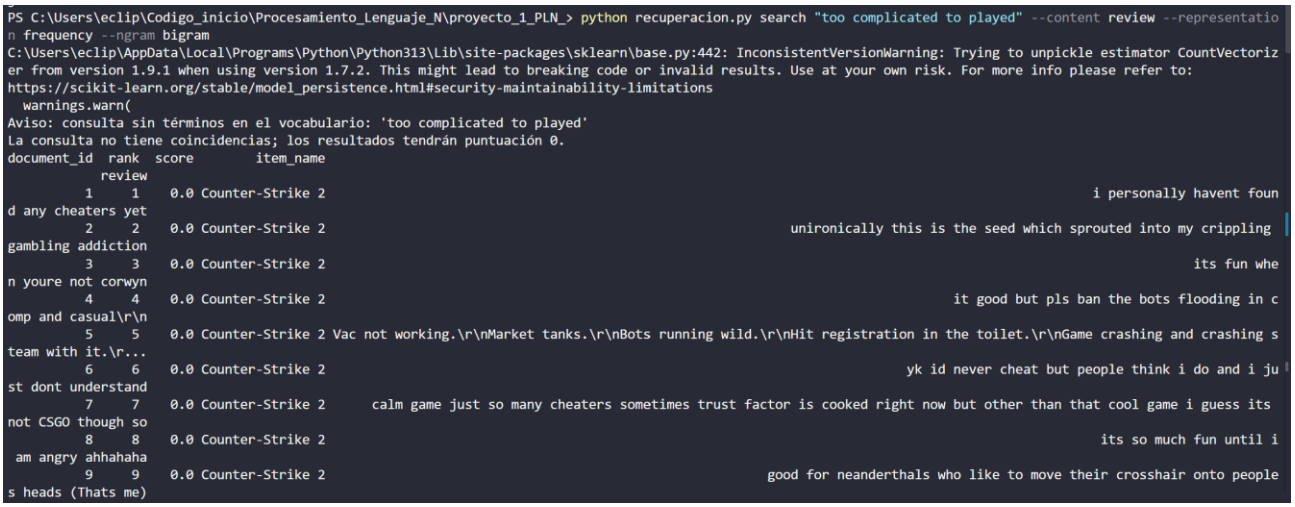
[Imagen 17: Resultados de "fantastic game" (binary, unigram, item_review)]

Poor Perfomance

1) **python recuperacion.py search "too complicated to played" --content review --representation frequency --ngram bigram**

Esta consulta solo arroja coincidencias con puntuación igual a 0 por lo que no trae resultados nuevos con lo que comparar, probablemente sea la forma en que está escrita, las palabras usadas o que sean más de dos palabras.

Después de una revisión parece ser que está mal la gramática, entonces ese puede ser un factor que afecte las consultas.

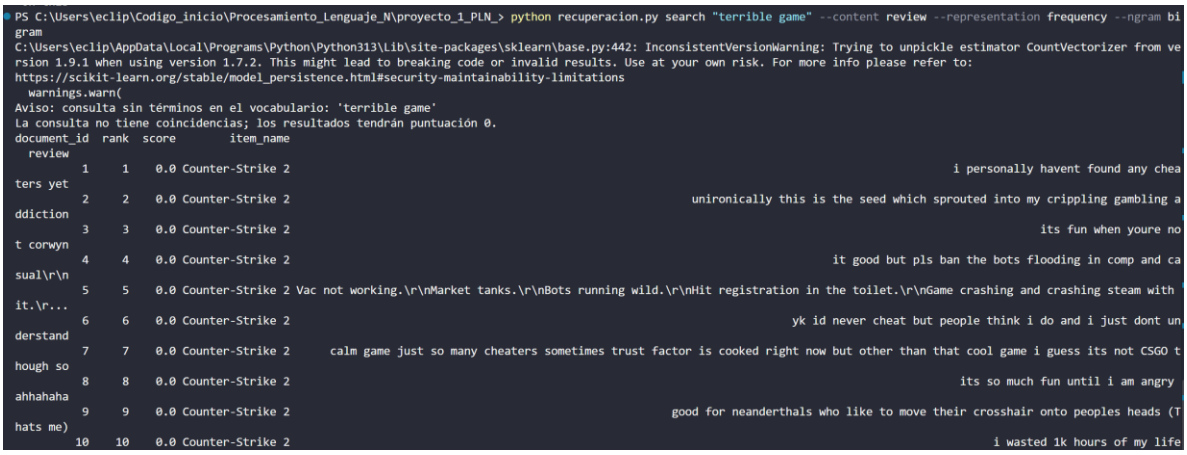


```
PS C:\Users\eclip\Codigo_inicio\Procesamiento_Lenguaje_N\proyecto_1_PLN_> python recuperacion.py search "too complicated to played" --content review --representatio
n frequency --ngram bigram
C:\Users\eclip\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\base.py:442: InconsistentVersionWarning: Trying to unpickle estimator CountVectoriz
er from version 1.9.1 when using version 1.7.2. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer to:
https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
  warnings.warn(
Aviso: consulta sin términos en el vocabulario: 'too complicated to played'
La consulta no tiene coincidencias; los resultados tendrán puntuación 0.
document_id  rank  score       item_name
           review
          1     1    0.0 Counter-Strike 2                                                                                                  i personally havent foun
d any cheaters yet
          2     2    0.0 Counter-Strike 2                                                                  unironically this is the seed which sprouted into my crippling
gambling addiction
          3     3    0.0 Counter-Strike 2                                                                                                                  its fun whe
n youre not corwyn
          4     4    0.0 Counter-Strike 2                                                                                        it good but pls ban the bots flooding in c
omp and casual\r\n
          5     5    0.0 Counter-Strike 2 Vac not working.\r\nMarket tanks.\r\nBots running wild.\r\nHit registration in the toilet.\r\nGame crashing and crashing s
team with it.\r\...
          6     6    0.0 Counter-Strike 2                                                                                                yk id never cheat but people think i do and i ju
st dont understand
          7     7    0.0 Counter-Strike 2      calm game just so many cheaters sometimes trust factor is cooked right now but other than that cool game i guess its
not CSGO though so
          8     8    0.0 Counter-Strike 2                                                                                                                  its so much fun until i
 am angry ahhahaha
          9     9    0.0 Counter-Strike 2                                                                  good for neanderthals who like to move their crosshair onto people
s heads (Thats me)
```

[Imagen 18: Resultados de "too complicated to played" con puntuación 0 (frequency, bigram)]

2)**python recuperacion.py search "terrible game" --content review --representation frequency --ngram bigram**

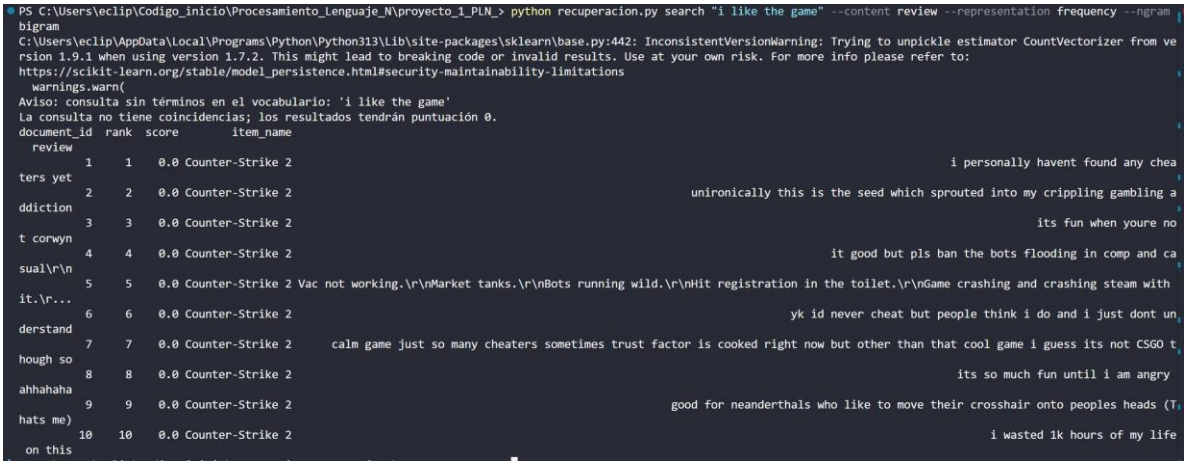
Para esta consulta se obtuvo resultados con puntuación igual a cero, por lo que al ser bigramas parece que presentan mayores problemas para encontrar palabras similares, incluso en contexto.



[Imagen 19: Resultados de "terrible game" con puntuación 0 (frequency, bigram)]

3) **python recuperacion.py search "i like the game" --content review --representation frequency --ngram bigram**

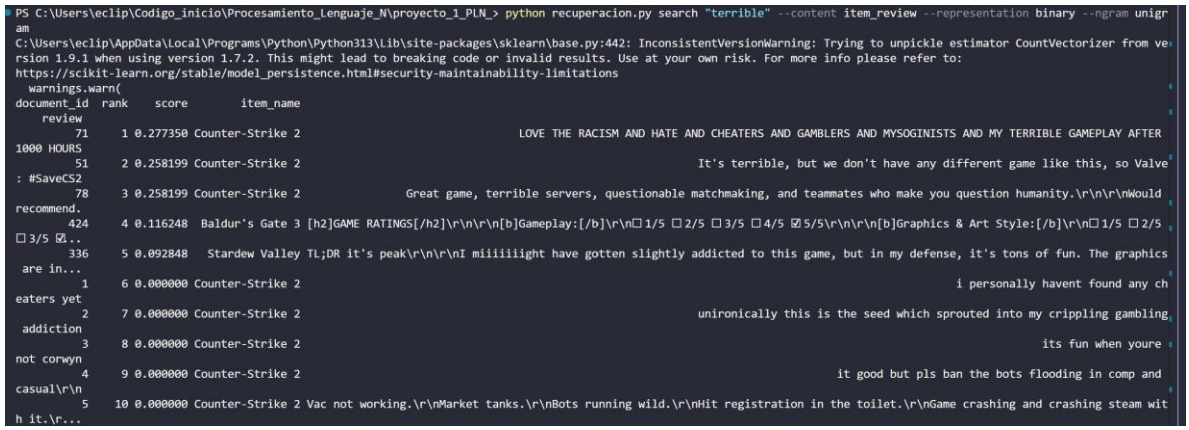
Para esta consulta obtuvimos resultados con puntuación igual a cero por lo que podemos confirmar que importa el número de palabras, que sean las palabras exactas para los bigramas.



[Imagen 16: Resultados de "i like the game" con puntuación 0 (frequency, bigram)]

4) **python recuperacion.py search "terrible" --content item_review --representation binary --ngram unigram**

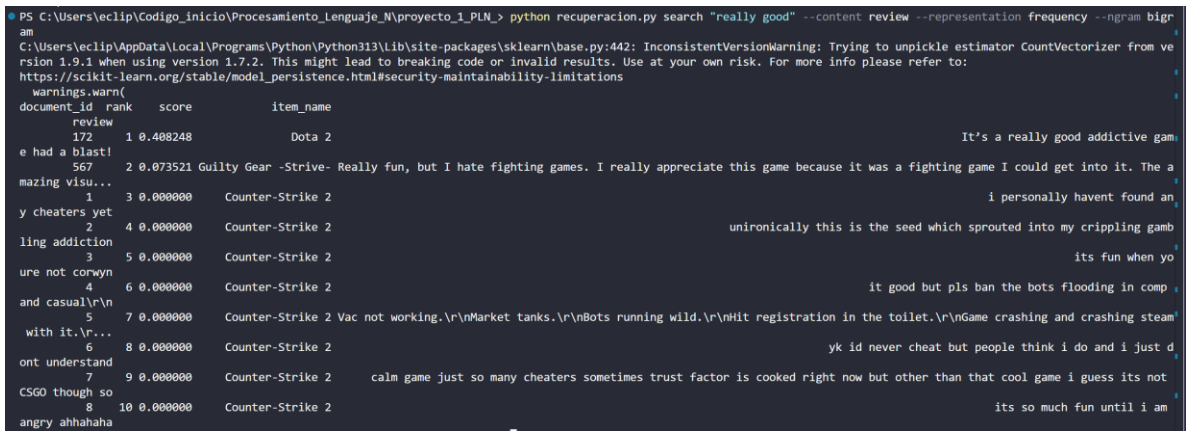
Para esta consulta obtuvimos pocos resultados que contengan la palabra, además de resultados con puntuación igual a cero, por lo que también puede influir la cantidad de comentarios ya que la palabra terrible es para un comentario negativo, estos pueden variar en cantidad dependiendo del juego.



[Imagen 20: Resultados de "terrible" con pocas coincidencias (binary, unigram, item_review)]

5) python recuperacion.py search "really good" --content review --representation frequency --ngram bigram

Para esta consulta a pesar de usar una palabra que normalmente tendría buenos resultados, como es un bigrama para ser que si no están en ese orden las mismas palabras o similares los resultados son pocos y también arroja con puntuación igual a 0.



[Imagen 21: Resultados de "really good" con pocas coincidencias (frequency, bigram)]